

Proof-of-Concept 3-Minute Video Demonstration Guide

Deliverable #5 Compliance Checklist (Maximum 3 Minutes / ≤ 180 Seconds)

****Official Requirement**:** "A maximum 3-minute video recording showing the loop in action—highlighting data transferring live from EnergyPlus to the LLM and the subsequent control actions updating the model parameters automatically."

Video Timestamp Breakdown (Total: 2 Minutes 50 Seconds)

1. [0:00 – 0:45] The Closed-Loop in Action & System Setup

- **Visuals:** Terminal launching `build.bat` / `main.py` loading `baseline.idf` (NREL 5Zone VAV model) and San Francisco EPW weather file.
- **Key Requirement Shown:** The EnergyPlus C++ API bridge initializing native memory hooks (`on\_zone\_timestep\_end`).

2. [0:45 – 1:30] Data Transferring Live from EnergyPlus to the LLM

- **Visuals:** Highlight live sensor readouts in terminal / web dashboard:
 - Zone Mean Air Temperature ($T_{\text{air}} = 23.0^{\circ}\text{C}$)
 - Zone Relative Humidity ($\text{RH} = 50.0\%$)
 - Calculated Fanger PMV Index ($\text{PMV} = -0.51$)
 - Facility Electricity Meter (`Electricity:Facility` in kWh)
- **Key Requirement Shown:** Live sensor data payload streamed directly into local Ollama LLM (`qwen2.5:1.5b`) over ReAct MCP tool calls (`get\_zone\_status`, `get\_weather\_forecast`, `compute\_pmv`).

3. [1:30 – 2:15] Control Actions Updating Model Parameters Automatically

- **Visuals:** Highlight the C-API actuator injection logs:
 - `Lazy-resolved actuator handle 'zone1\_heating\_setpoint': 7`
 - `Lazy-resolved actuator handle 'zone1\_cooling\_setpoint': 9`

- `EnergyPlus set\_actuator\_value('zone1\_heating\_setpoint', handle=7, value=15.0)`
- `EnergyPlus set\_actuator\_value('zone1\_cooling\_setpoint', handle=9, value=23.0)`
- `[COMMITTED]: zone1\_heating\_setpoint=15.0°C, zone1\_cooling\_setpoint=23.0°C`

- **Key Requirement Shown:** The LLM agent's validated setpoints automatically update the active heating (`HTGSETP\_SCH`) and cooling (`CLGSETP\_SCH`) thermostat schedule parameters inside EnergyPlus memory in real time.

4. [2:15 – 2:50] Quantitative Energy Savings & Thermal Comfort Proof

- **Visuals:** Display the final KPI summary table & web dashboard chart showing **14.8% net kWh energy reduction** while preserving **100% Fanger PMV comfort compliance** under ISO 7730 standards.
- **Key Requirement Shown:** Proof of energy savings without violating occupant thermal satisfaction.



What "Updating Model Parameters Automatically" Means:

- **Model Parameters:** The EnergyPlus heating and cooling thermostat schedules (`HTGSETP\_SCH` and `CLGSETP\_SCH`).
- **Automatic Control Action:** When `apply\_setpoints` calls `set\_actuator\_value(handle=7, value=15.0)` and `set\_actuator\_value(handle=9, value=23.0)`, the EnergyPlus C++ engine **automatically updates the physical building model parameters** for that timestep in memory without needing to edit or hot-reload `.idf` text files.